

# DRAFT California State Water Resources Control Board’s Guide to Publishing on the California Open Data Portal

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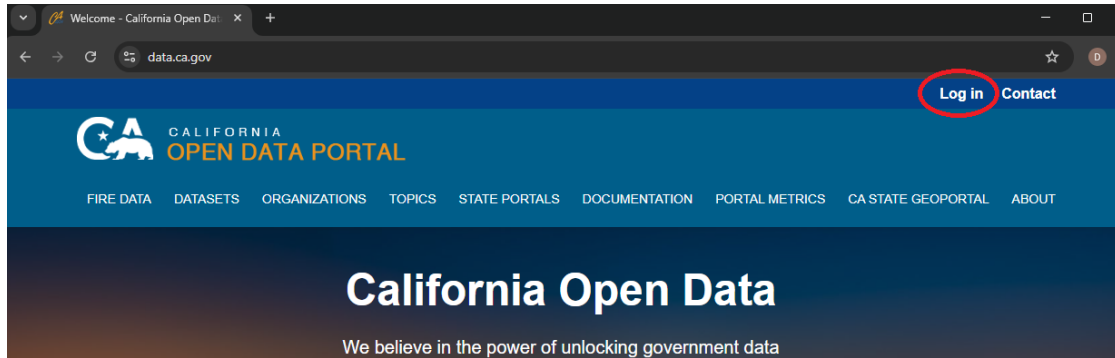
## 1 Introduction

This document describes the basics of how to publish and maintain a dataset on the [California Open Data Portal](#) for staff at the California State Water Resources Control Board. In the context of the California Open Data Portal, a dataset is a collection of data resources (which are files or links to files containing data, such as CSV files), together with a description and other supporting information (metadata), at a fixed URL.

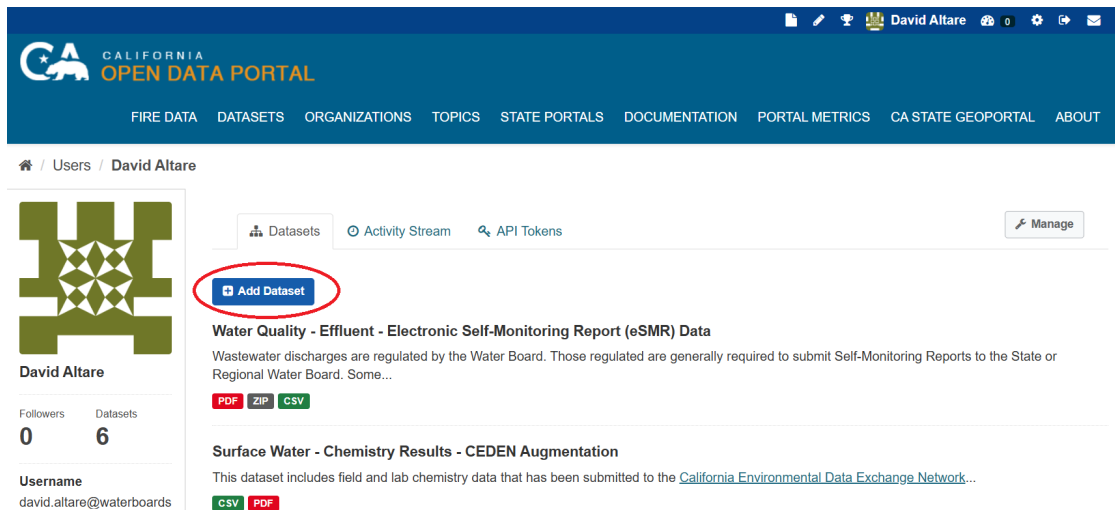
The instructions below demonstrate how to create a new dataset, then add one or more data resources to that dataset. If you want to manage an existing dataset (including adding a new data resource, updating an existing data resource, or editing dataset metadata), skip to the [Managing an Existing Dataset or Data Resource](#) section.

## 2 Creating a New Dataset

1. Go to <https://data.ca.gov> and click on the **Log in** button at the top right of the window to log in to your user account. (NOTE: if you don't already have an account on the data.ca.gov portal, you'll need to contact Office of Information Management and Analysis (OIMA) staff to create an account and get added to the Water Board's organizational account.)



2. In your account dashboard, click on the **Add Dataset** button.



3. Fill in the dataset metadata (e.g., Title, Description, etc.). Guidance on how to populate the metadata is available [here](#) and [here](#).

The screenshot shows the 'Create Dataset' page on the California Open Data Portal. The browser address bar shows the URL: `data.ca.gov/dataset/new?group=a652035f-505c-4c24-8464-fed3623acfd`. The page header includes the California Open Data Portal logo and navigation links: FIRE DATA, DATASETS, ORGANIZATIONS, TOPICS, STATE PORTALS, DOCUMENTATION, PORTAL METRICS, CA STATE GEOPORTAL, and ABOUT. The breadcrumb trail is: Home / Datasets / Create Dataset.

**What are datasets?**

A CKAN Dataset is a collection of data resources (such as files), together with a description and other information, at a fixed URL. Datasets are what users see when searching for data.

**1 Create dataset** **2 Add data**

**\* Title:**

eg. A descriptive title

**\* URL:** `data.ca.gov/dataset/<dataset>` [Edit](#)

**\* Description:**

eg. Some useful notes about the data

You can use [Markdown formatting here](#)

NOTE: Use the **Visibility** dropdown menu to control whether or not your dataset is publicly accessible. Use the **Private** option to restrict visibility of your dataset to only other users of the Water Board's organizational account on the Open Data Portal (this is useful when drafting your dataset prior to publication), or use the **Public** option to make your dataset accessible to the general public.

**Topic :**

For data curation purposes users should be a member of a topic to add a dataset to the topic

**\* Publisher:**

**Visibility:**

Private

Private

Public

Public

**\* Rights:**

**\* Program Contact Name:**

When you're finished, click on the **Next: Add Data** button at the bottom of the page.

Custom Field:

Key:  Value:

Custom Field:

Key:  Value:

Custom Field:

Key:  Value:

The data license you select above only applies to the contents of any resource files that you add to this dataset. By submitting this form, you agree to release the metadata values that you enter into the form under the [Open Database License](#).

\* Required field

**Next: Add Data**

4. Add a file containing your data using either the **Upload** or **Link** buttons, add a title and description for the resource, then click the **Finish** button, or click the **Save & add another** button to add another data resource. Guidance on how to format a data file for publication on the open data portal is available [here](#) and [here](#).

1 Create dataset 2 Add data

**File:**

**Upload** **Link**

**Title:**

eg. January 2015 Gold Prices

**Description:**

Some useful notes about the data

You can use Markdown formatting here

**Format:**

eg. CSV, XML or JSON

Previous **Save & add another** **Finish**

5. Add a data dictionary for the data resource you just created using one or both of the following methods:

- A: Upload a PDF containing the data dictionary as a separate resource in the dataset
- B: Add the data dictionary directly to the data resource's page (the data dictionary will appear below the data table)

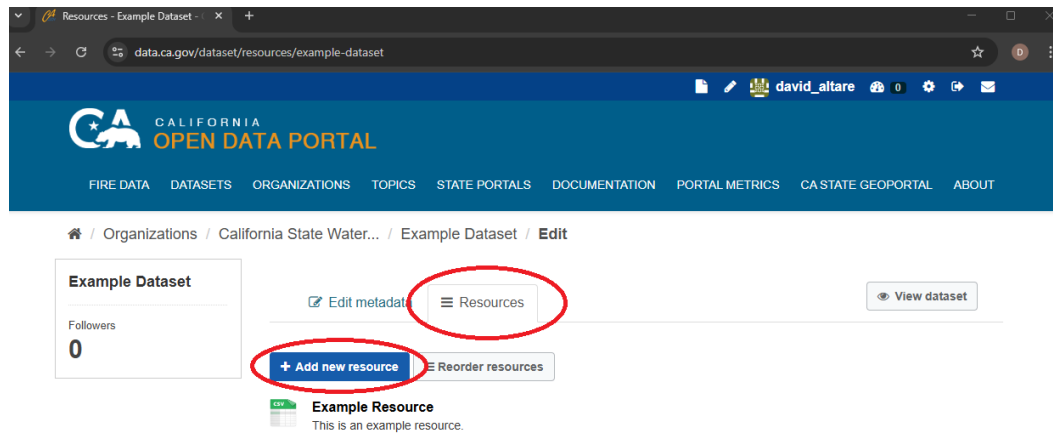
Guidance on how to populate the data dictionary is available [here](#), [here](#), and [here](#).

A. Upload a PDF containing the data dictionary to the dataset (as a separate resource), by going to the dataset homepage and clicking on the **Manage** button.

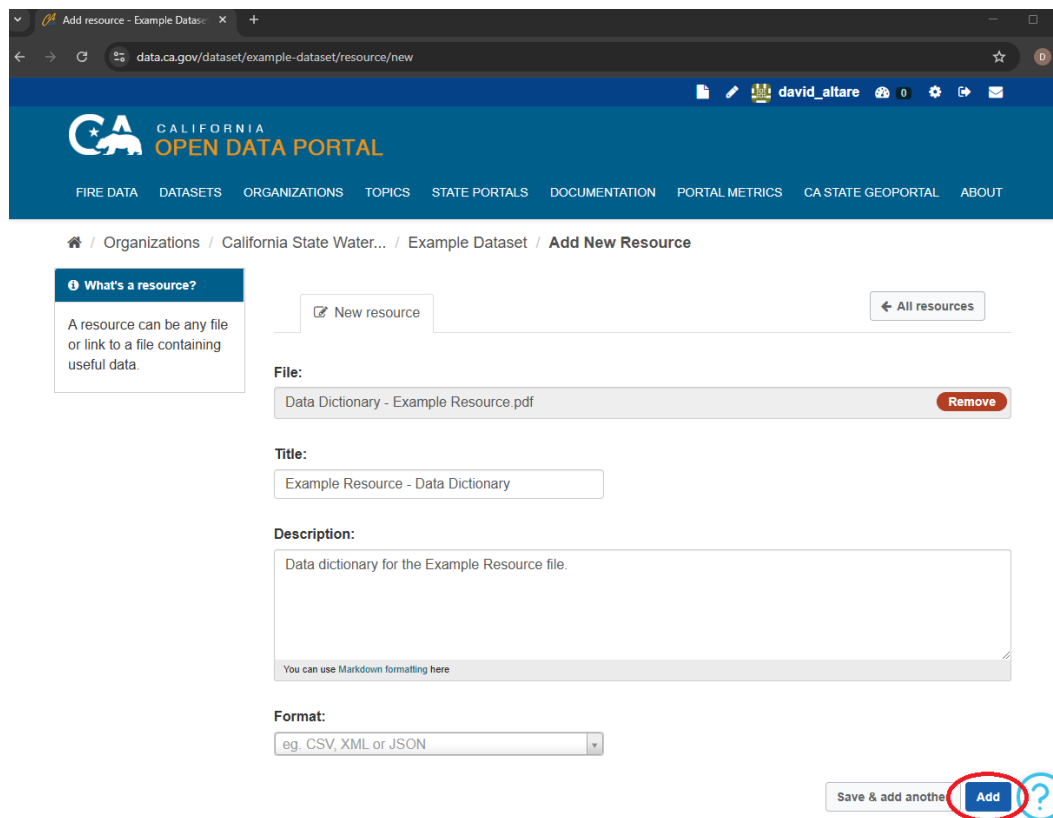
The screenshot shows the 'Example Dataset' page on the California Open Data Portal. The page is titled 'Example Dataset' and is marked as 'PRIVATE'. It features a 'Manage' button circled in red in the top right corner. The page includes a 'Data and Resources' section with an 'Example Resource' and a search bar containing the word 'water'. Below this is an 'Additional Info' section with a 'Download' button and a table of metadata.

| Field                 | Value                         |
|-----------------------|-------------------------------|
| Public Access Level   | Public                        |
| Rights                | No restrictions on public use |
| Program Contact Name  | Contact Name                  |
| Program Contact Email | waterdata@waterboards.ca.gov  |

Then click on the **Resources** tab, and click on the **Add new resource** button.



Upload the PDF file by clicking the **Upload** button, then click on the **Add** button.



- B. Add the data dictionary directly to the data resource by going to the data resource's page and clicking on the **Manage** button. (**NOTE:** After uploading a new data resource, before you can add data dictionary information you'll need to wait until the data table appears on the resource's page, as shown in the screenshot below; this could take a few minutes, depending on the size of the resource.)

The screenshot shows the 'Example Resource' page on the California Open Data Portal. The page has a blue header with the portal logo and navigation links. Below the header, the breadcrumb trail is 'Organizations / California State Water... / Example Dataset / Example Resource'. The main content area features a title 'Example Resource' and a row of action buttons: 'Setup for Reporting', 'DataStore', 'Manage' (circled in red), 'Download', and 'Data API'. Below these buttons, the URL is displayed: 'https://data.ca.gov/dataset/862443bb-e5c3-4ee0-9bac-44771c406b09/resource/194a9af0-4111-4472-a64c-b891b0652f2f/download/penguins...'. A note states 'This is an example resource.' Below this, there are buttons for 'Data Table' and 'Canonical View'. Further down, there are buttons for 'Fullscreen' and 'Embed'. A section titled 'Add Filter' is visible, followed by a 'Show' dropdown set to '10' and a 'Hide/Unhide Columns' button. A search bar is also present. At the bottom, a table displays the first three entries of the dataset.

| _id | species | island    | bill_length_mm | bill_depth_mm | flipper_length_mm | body_mass_g | sex    | year |
|-----|---------|-----------|----------------|---------------|-------------------|-------------|--------|------|
| 1   | Adelie  | Torgersen | 39.1           | 18.7          | 181               | 3750        | male   | 2007 |
| 2   | Adelie  | Torgersen | 39.5           | 17.4          | 186               | 3800        | female | 2007 |
| 3   | Adelie  | Torgersen | 40.3           | 18            | 195               | 3250        | female | 2007 |

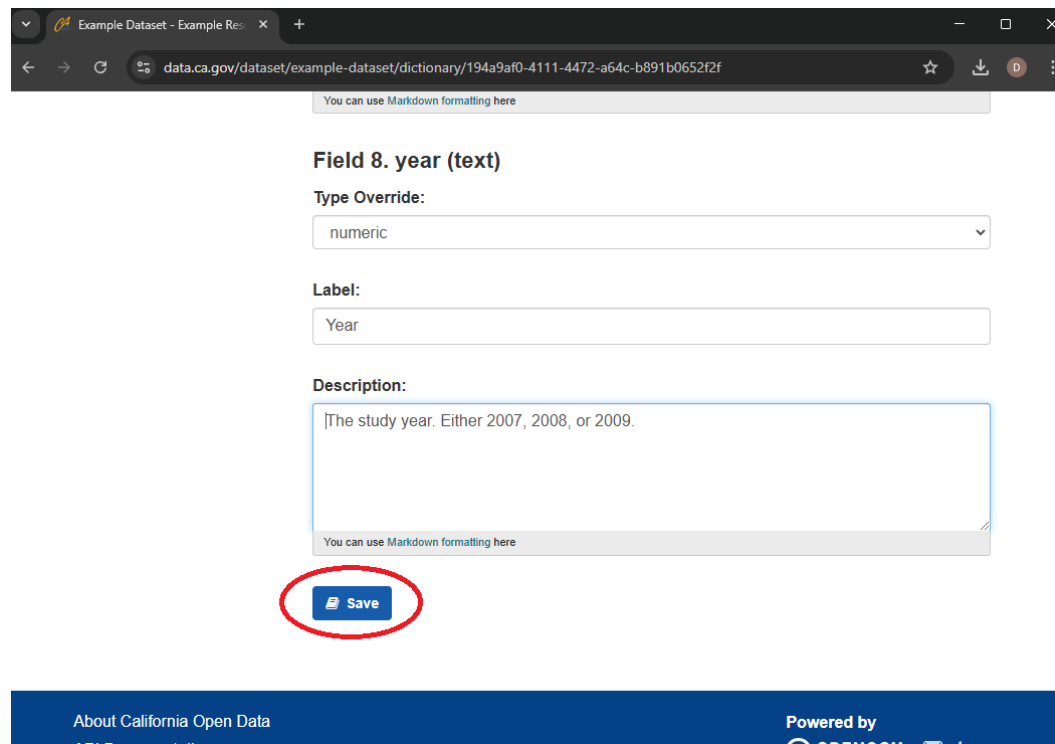


Go to the **Data Dictionary** tab, then fill in the available information for each field in the data resource.

The screenshot shows a web browser window with the URL `data.ca.gov/dataset/example-dataset/dictionary/194a9af0-4111-4472-a64c-b891b0652f2f`. The page header includes the California Open Data Portal logo and navigation links: FIRE DATA, DATASETS, ORGANIZATIONS, TOPICS, STATE PORTALS, DOCUMENTATION, PORTAL METRICS, CA STATE GEOPORTAL, and ABOUT. The breadcrumb trail is: Home / Organizations / California State Water... / Example Dataset / Example Resource / Edit. On the left, a sidebar shows 'Example Resource' with a 'Format' dropdown set to 'CSV'. The main content area has tabs: 'Edit resource', 'DataStore', 'Data Dictionary' (highlighted with a red circle), 'Setup for Reporting', and 'Views'. Above the 'Data Dictionary' tab are buttons for 'All resources' and 'View resource'. The 'Data Dictionary' tab contains the following fields:

- Field 1. species (text)**
- Type Override:** A dropdown menu.
- Label:** A text input field.
- Description:** A large text area with a note at the bottom: 'You can use Markdown formatting here'.

Click the **Save** button when finished.



The screenshot shows a web browser window with the URL `data.ca.gov/dataset/example-dataset/dictionary/194a9af0-4111-4472-a64c-b891b0652f2f`. The page displays a form for editing a dataset field. At the top, a message states "You can use Markdown formatting here". The form includes the following sections:

- Field 8. year (text)**
- Type Override:** A dropdown menu currently set to "numeric".
- Label:** A text input field containing the word "Year".
- Description:** A text area containing the text "The study year. Either 2007, 2008, or 2009." Below this text area is another "You can use Markdown formatting here" message.

At the bottom of the form, a blue button with a floppy disk icon and the text "Save" is circled in red. The footer of the page is a dark blue bar containing the text "About California Open Data" and "Powered by" followed by the "OPENGOV" logo.

After saving the data dictionary information, the data dictionary will appear below the data table on the resource's page. (**NOTE:** you may need to update the resource's **DataStore** to make sure all of the changes to the data dictionary are applied to the resource; to do this, see the instructions in [Step 2](#) of the *Managing an Existing Dataset or Data Resource* section below.)

| _id | species | island    | bill_length_mm | bill_depth_mm | flipper_length_mm | body_mass_g | sex    | year |
|-----|---------|-----------|----------------|---------------|-------------------|-------------|--------|------|
| 1   | Adelie  | Torgersen | 39.1           | 18.7          | 181               | 3750        | male   | 2007 |
| 2   | Adelie  | Torgersen | 39.5           | 17.4          | 186               | 3800        | female | 2007 |
| 3   | Adelie  | Torgersen | 40.3           | 18            | 195               | 3250        | female | 2007 |
| 4   | Adelie  | Torgersen | NaN            | NaN           | NaN               | NaN         | NA     | 2007 |
| 5   | Adelie  | Torgersen | 36.7           | 19.3          | 193               | 3450        | female | 2007 |
| 6   | Adelie  | Torgersen | 39.3           | 20.6          | 190               | 3650        | male   | 2007 |
| 7   | Adelie  | Torgersen | 38.9           | 17.8          | 181               | 3625        | female | 2007 |
| 8   | Adelie  | Torgersen | 39.2           | 19.6          | 195               | 4675        | male   | 2007 |
| 9   | Adelie  | Torgersen | 34.1           | 18.1          | 193               | 3475        | NA     | 2007 |
| 10  | Adelie  | Torgersen | 42             | 20.2          | 190               | 4250        | NA     | 2007 |

Showing 1 to 10 of 344 entries

| Column         | Type    | Label            | Description                                                                                                      |
|----------------|---------|------------------|------------------------------------------------------------------------------------------------------------------|
| species        | text    | Species          | The penguin's species. Either Adelie, Chinstrap, or Gentoo.                                                      |
| island         | text    | Island           | The island in Palmer Archipelago, Antarctica where the penguin was observed. Either Biscoe, Dream, or Torgersen. |
| bill_length_mm | numeric | Bill Length (mm) | The penguin's bill length in millimeters.                                                                        |
| bill_depth_mm  | numeric | Bill Depth (mm)  | The penguin's bill depth in millimeters.                                                                         |

**NOTE:** Setting the data type for each field by selecting a value in the **Type Override** box is especially useful if your users will be accessing the data via the API, but you'll need to ensure that all of the data in the given field is compatible with that data type before setting a data type other than 'text'. If there are problems with the data types, you can find more information by following the instructions in [Step 2](#) of the *Managing an Existing Dataset or Data Resource* section below, and going to the **DataStore** tab.

If you need to add data dictionary information for data resources with many fields and/or datasets with large numbers of resources, you can contact OIMA staff for assistance with automating the process of uploading that data dictionary information (to avoid the need for manual entry of the data dictionary information).

### 3 Managing an Existing Dataset or Data Resource

#### 3.1 Update Dataset Metadata or Add Additional Data Resource

To add an additional data resource to an existing dataset, or to update the metadata for an existing dataset:

1. Go to the dataset and click on the **Manage** button.

The screenshot shows the California Open Data Portal interface. The top navigation bar includes links for FIRE DATA, DATASETS, ORGANIZATIONS, TOPICS, STATE PORTALS, DOCUMENTATION, PORTAL METRICS, CA STATE GEOPORTAL, and ABOUT. The user is logged in as 'david\_altare'. The breadcrumb trail shows the path: Home / Organizations / California State Water... / Example Dataset. The main content area is titled 'Example Dataset' and includes a 'Test description' section. Below this is the 'Data and Resources' section, which lists two resources: 'Example Resource' (a CSV file) and 'Example Resource - Data Dictionary' (a PDF file). Each resource has an 'Explore' button. A search bar with the text 'water' is visible. At the bottom, there is an 'Additional Info' section with a 'Download' button. The 'Manage' button, located in the top right corner of the dataset page, is circled in red.

Example Dataset

Followers: 0

Follow

Organization: California State Water Resources Control Board

Website: <https://www.waterboards.ca.gov/> To preserve, enhance, and restore the state's water resources.

Dataset Topics Showcases Activity Stream Manage

Example Dataset

Test description

Data and Resources

Example Resource: This is an example resource. Explore

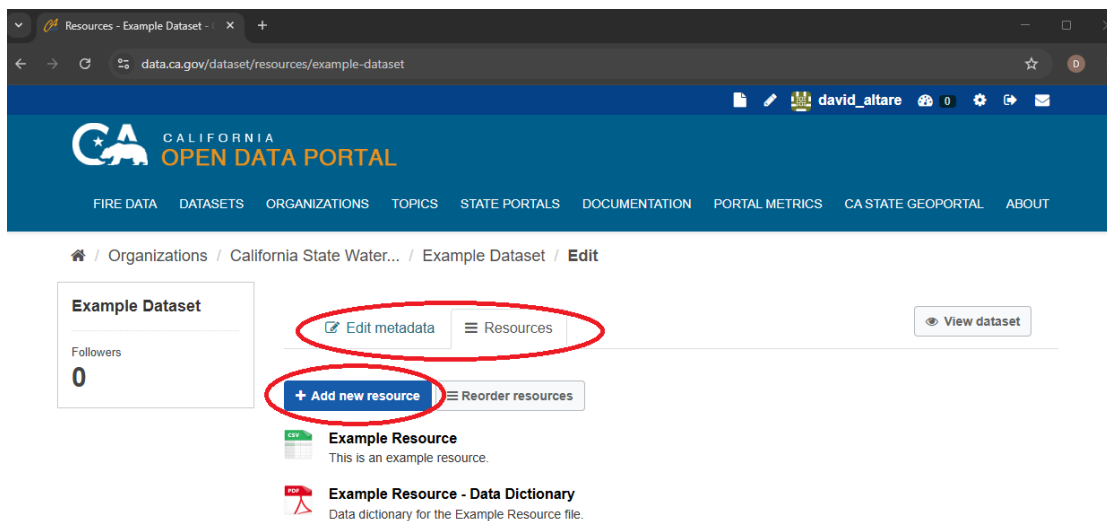
Example Resource - Data Dictionary: Data dictionary for the Example Resource file. Explore

water

Additional Info Download

| Field | Value |
|-------|-------|
|-------|-------|

2. To update the dataset's metadata, click on the **Edit metadata** tab. To add a new data resource to the dataset click on the **Resources** tab then click on the **Add new resource** button.



If you're adding a new data resource that doesn't already have an associated data dictionary available on the portal, you'll need to add a data dictionary for your resource. See [Step #6](#) above for information on how to do this.

## 3.2 Manage Existing Data Resource

To manage an existing data resource (i.e., an existing file within a dataset and/or the file's associated metadata):

1. Go to the data resource's page, then click on the **Manage** button.

The screenshot shows the 'Example Resource' page on the California Open Data Portal. The page has a blue header with the portal's logo and navigation links. Below the header, the breadcrumb trail reads: Home / Organizations / California State Water... / Example Dataset / **Example Resource**. The main content area features a title 'Example Resource' and a row of action buttons: 'Setup for Reporting', 'DataStore', 'Manage' (circled in red), 'Download', and 'Data API'. Below these buttons, the URL is displayed: <https://data.ca.gov/dataset/862443bb-e5c3-4ee0-9bac-44771c406b09/resource/194a9af0-4111-4472-a64c-b891b0652f2f/download/penguins...>. A message states 'This is an example resource.' Below this, there are tabs for 'Data Table' (selected) and 'Canonical View'. To the right of the 'Data Table' tab are buttons for 'Fullscreen' and 'Embed'. Further down, there is an 'Add Filter' button, a 'Show 10 entries' dropdown, and a 'Hide/Unhide Columns' button. A search bar is also present. At the bottom, a table displays data for penguins.

| _id | species | island    | bill_length_mm | bill_depth_mm | flipper_length_mm | body_mass_g | sex    | year |
|-----|---------|-----------|----------------|---------------|-------------------|-------------|--------|------|
| 1   | Adelle  | Torgersen | 39.1           | 18.7          | 181               | 3750        | male   | 2007 |
| 2   | Adelle  | Torgersen | 39.5           | 17.4          | 186               | 3800        | female | 2007 |

2. Go to the **Edit Resource** tab to update the data file or update the resource's title or description, go to the **DataStore** tab to ensure that the data has been correctly uploaded to the database that feeds the resource's API, or go to the **Data Dictionary** tab to add information about each field and set the fields' data types.

data.ca.gov/dataset/example-dataset/resource/194a9af0-4111-4472-a64c-b891b0652f2f/edit

CA CALIFORNIA OPEN DATA PORTAL

FIRE DATA DATASETS ORGANIZATIONS TOPICS STATE PORTALS DOCUMENTATION PORTAL METRICS CA STATE GEOPORTAL ABOUT

Organizations / California State Water... / Example Dataset / Example Resource / Edit

Example Resource

Format CSV

All resources View resource

Edit resource DataStore Data Dictionary Setup for Reporting Views

File:

penguins.csv Remove

Title:

Example Resource

Description:

This is an example resource.

You can use Markdown formatting here

Format:

CSV

Delete Update Resource ?

## 4 Additional Resources

For more information about publishing open data resources on the California Open Data Portal, see the following resources:

- [California Open Data Publisher's Handbook](#)
- [California Open Data Portal Publisher's Guide](#)
- [California State Water Board's Open Data Publishing Guide](#)
- [California State Water Board's Recommendations and Guidelines for Data Dictionary Development](#)
- [California State Water Board's Open Data Handbook](#)